# LTA Exact Maximum-Symmetry Search Acceleration Specification

Designed for package v7.11 (MS6)

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**Current authority routing:** supported Group-I/II species, exact integer populations, formal charges, and charge neutrality are owned by `doc/specs/v7_13_group_i_ii_cation_extension.md`; the Al24Si24O96 charge-equivalent dry-volume model and empirical endpoint provenance are owned by `doc/specs/v7_14_group_i_ii_volume_reference_extension.md`. This specification imports those semantics where chemistry or volume enters its surface.

## 1. Purpose

MS6 accelerates the exact-coordinate optimizer without weakening its mathematical guarantees. It addresses the large low-symmetry search cost found during the v7.10 extensive validation.

The implementation adds:

1. a proof-producing forced-variant fast path;
2. one reusable exact-realization catalog per tested subgroup;
3. explicit primary-order and tie-search accounting;
4. richer subgroup and occupied-orbit diagnostics;

5. LTA-capacity regression fixtures with 24 four-ring, 16 six-ring, and 6 eight-ring sites.

MS6 does not change the optimization objective, accepted variant-library policies, numerical tolerances, or post-write certification rules.

# 2. Observed bottleneck

For a candidate subgroup $H$, the MS3 implementation examined many subgroup-constant occupation colorings. For every coloring it repeated the same representative-variant tests on each occupied $H$-orbit.

The orbit-level realization problem depends on

$$(H, O, \mathcal{V}, A),$$

where $O$ is a physical-site orbit, $\mathcal{V}$ is the active variant domain, and $A$ is the variant transition relation. It does not depend on which member of `SUPPORTED_CATIONS` labels $O$.

The repeated tests were therefore redundant. A second special case was more severe: when every site has exactly one parent-equivariant active variant, the coordinate assignment is forced, but MS3 still traversed the variant-combination search.

# 3. Preserved optimization contract

The exact objective remains lexicographic:

$$\max |H_{\mathrm{coord}}|, \qquad \text{then} \qquad \max |H_{\mathrm{occ}}|.$$

Every returned solution must satisfy

$$H_{\mathrm{coord}} \le H_{\mathrm{occ}} \le G,$$

and the certificate must prove that every subgroup order larger than $|H_{\mathrm{coord}}|$ is infeasible or absent.

MS6 may skip work only when a theorem proves that the skipped search cannot change the answer.

# 4. Forced-variant fast path

## 4.1 Predicate

Let $V_s$ be the active variants owned by physical site $s$. The fast path is enabled only when

$$|V_s| = 1 \qquad \forall s \in \mathcal{P},$$

and every parent operation $g \in G$ has a unique transition from the sole source variant to the sole variant at the mapped destination site:

$$A_g(s, v_s) = (g(s), v_{g(s)}).$$

A missing or ambiguous transition disables the fast path. Merely having one listed variant per site is insufficient.

## 4.2 Theorem

Under the predicate above, every grouped-site occupation $c$ has one coordinate realization $X(c)$ and

$$\mathrm{Stab}_G(X(c)) = \mathrm{Stab}_G(c).$$

Proof: an operation preserves $X(c)$ only if it preserves the species label at every mapped site, so $H_{\mathrm{coord}} \le H_{\mathrm{occ}}$. Conversely, if it preserves $c$, the unique transition condition maps every occupied site's sole variant to the sole variant at the destination, so $H_{\mathrm{occ}} \le H_{\mathrm{coord}}$.

Therefore MS6 may call the MS2 occupation optimizer and lift its certified result directly to an exact coordinate result.

### 4.3 Failure safety

The lift recomputes both stabilizers geometrically. It raises `ExactCoordinateOptimizationError` unless

$$H_{\text{coord}} = H_{\text{occ}}.$$

The variant-search node count is zero. The certificate records:

```
exact_search_strategy = forced_variant_lift
forced_variant_fast_path = true
```

## 5. Subgroup-orbit realization catalog

### 5.1 Data structures

`OrbitVariantRealizationCatalogEntry` stores, for one subgroup orbit:

```
site_orbit
status: feasible | infeasible | indeterminate
choices: tuple[OrbitVariantRealization, ...]
representative test counters
```

`SubgroupVariantRealizationCatalog` stores all entries for one subgroup and aggregate diagnostic counters.

### 5.2 Construction

For each orbit $O$ of subgroup $H$:

1. choose the canonical representative site $s = \min O$;
2. test every active variant owned by $s$;
3. retain each variant whose propagated assignment is complete, unique, and composition-consistent;
4. classify the orbit as:
   - feasible if at least one realization exists;
   - indeterminate if no realization exists but at least one required transition is ambiguous;
   - infeasible otherwise.

The catalog is built once for every subgroup representative tested by the optimizer.

### 5.3 Reuse

For a subgroup-constant occupation, exact feasibility is obtained by selecting the catalog entries for occupied orbits. Empty orbits require no variant realization.

A reused catalog performs no additional representative tests and consumes no additional `VariantRealizationBudget` nodes.

## 6. Primary certification and tie search

The subgroup loop is ordered by descending subgroup order. The first verified exact candidate at order $q$ proves the primary optimum after all larger orders have been exhausted.

MS6 records this logical boundary explicitly:

```
primary_order_certified_before_tie_search = true
```

The optimizer may then continue only at order $q$ to satisfy the existing secondary occupation-symmetry and bounded tie contracts. It never visits lower subgroup orders after $q$ is known.

The general multi-variant path remains exact and retains the existing ambiguity and incomplete-search failures.

## 7. Certificate extensions

`CoordinateOptimalityCertificate` adds:

```
exact_search_strategy
forced_variant_fast_path
variant_orbit_catalogs_built
variant_orbit_catalog_cache_hits
primary_order_certified_before_tie_search
```

The two implemented strategies are:

```
forced_variant_lift
catalogued_subgroup_variant_search
```

`variant_orbit_catalog_cache_hits` counts occupation-feasibility analyses served from an already built subgroup catalog.

## 8. Diagnostic reporting

The maximum-symmetry report now includes:

- exact search strategy;
- forced-fast-path status;
- catalog build/reuse counts;
- parent-conjugacy key of the coordinate stabilizer;
- deterministic subgroup generators;
- finite order of every preserved operation;
- affine rotation determinant, trace, and fractional translation;
- occupied coordinate-subgroup orbit counts and sizes by ring family and species.

These diagnostics supplement, but do not replace, the authoritative operation-index stabilizer.

## 9. Public API

New public types and function:

```
OrbitVariantRealizationCatalogEntry
SubgroupVariantRealizationCatalog
build_subgroup_variant_realization_catalog(...)
```

`analyze_exact_variant_feasibility(...)` accepts an optional `catalog`. Omitting it preserves the prior standalone behavior.

`optimize_maximum_symmetry_coordinates(...)` keeps its existing signature. Strategy selection is automatic and deterministic.

## 10. Invariants

1. The fast path is used only for a unique total parent-equivariant variant assignment.
2. Fast-path results satisfy $H_{\mathrm{coord}} = H_{\mathrm{occ}}$ by direct recomputation.
3. Catalog entries are species-independent.
4. A catalog belongs to exactly one subgroup operation set.
5. Reusing a catalog does not consume variant-search nodes.
6. Ambiguous orbit entries remain indeterminate and cannot certify infeasibility.
7. Search guards still return no partial optimum.
8. Existing random, unique-batch, serialization, and smoothing behavior is unchanged.

## 11. Focused verification

MS6 tests must include:

1. LTA-capacity order-2 and order-1 exact cases with `max_variant_search_nodes=1`;
2. zero variant-search nodes on the forced path;
3. equality of coordinate and occupation stabilizers on the forced path;
4. deterministic repeated scale-fixture results;
5. direct catalog reuse with unchanged budget count;
6. a multi-variant exact search using the catalogued strategy;
7. report-level subgroup and occupied-orbit diagnostics;
8. CIF round-trip certification of an LTA-capacity low-symmetry result;
9. complete regression of MS0-MS5 and legacy workflows.

## 12. Algorithmic provenance

The finite group-action, subgroup, orbit, and stabilizer framework follows the references already cited in the architecture manual: Hart–Forcade, SOD, Lian *et al.*, Seress, and WYCKSPLIT.

The forced-variant theorem, subgroup-orbit realization catalog, certificate fields, and acceleration policy are package-specific consequences of that framework. The catalog uses ordinary memoization of a species-independent subproblem; no external enumeration algorithm is copied.